

CRF: The Complete Algebraic Structure

K20 from Maximum Entropy · K14 Irreducible · Theory Complete

All algebraic gaps closed or proved transcendently irreducible. One empirical item remains.

Ougit Jarittum

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Abstract

This paper closes the last two algebraic open questions from State Summary v15.

Theorem 3 (K20 from Maximum Entropy): The CRF inter-event time distribution maximises entropy subject to a fixed mean T_{avg} (Jaynes 1957; physically grounded in #21b). The maximum-entropy distribution for a positive random variable with fixed mean is exponential. For the exponential distribution, $P(T > T_{\text{avg}}) = e^{-1}$. The wave amplitude F equals the KL-action per cycle times this survivor fraction: $F = K^*T_{\text{avg}} \cdot e^{-1}$, hence $K^*T_{\text{avg}}/F = e$.

K14 irreducibility proof: For $F\varphi^2 = 4$ to hold exactly, ψ_{grad} would need to equal $(227 - 95\sqrt{5})/(285 + 95\sqrt{5})$ — an algebraic number. But $\psi_{\text{grad}} = \varepsilon_0(n+2)/(\sqrt{2n+3} \ln \varphi)$ involves $\ln \varphi$, which is transcendental. By Lindemann–Weierstrass, a transcendental number cannot equal an algebraic number. The 0.057% gap in K14 is irreducible.

Status: Every algebraic gap in CRF is now either zero, algebraically exact, or proved transcendently irreducible. The only remaining open item is P0-H Subject 4 (empirical gate).

1 Theorem 3: K20 from Maximum Entropy

Theorem 1 (K20 from MaxEnt). $K^*T_{\text{avg}}/F = e$.

Proof. Step 1 (maximum entropy of inter-event times): From instability #21b (V20.3, 4613 events), $H_{\text{before}} = 1.764 \approx 0.85H_{\text{after}}$. The entropy between R-events does not decay to zero: ε_0 noise continuously injects entropy. Between each R-event the system operates near maximum uncertainty. The inter-event time T has support on $[0, \infty)$ with fixed mean T_{avg} (from Wald, #21a).

Step 2 (Jaynes maximum entropy theorem): Among all probability distributions on $[0, \infty)$ with fixed mean μ , the unique maximum-entropy distribution is the exponential:

$$T \sim \text{Exp}(1/\mu) \quad \Leftrightarrow \quad P(T > t) = e^{-t/\mu}$$

(Jaynes, 1957; information-theoretic derivation of the canonical distribution.)

Step 3 (survivor fraction at the mean): For $T \sim \text{Exp}(1/T_{\text{avg}})$:

$$P(T > T_{\text{avg}}) = e^{-T_{\text{avg}}/T_{\text{avg}}} = e^{-1} = \frac{1}{e}$$

Step 4 (wave amplitude definition): The wave amplitude F is the amplification of the IDV transfer per R-event. It equals the KL-action per cycle (K^*T_{avg} , the threshold accumulated over one mean inter-event period) weighted by the probability that a node has *not* yet fired by that time:

$$F = K^*T_{\text{avg}} \cdot P(T > T_{\text{avg}}) = K^*T_{\text{avg}} \cdot \frac{1}{e}$$

Rearranging: $K^*T_{\text{avg}}/F = e$ □

Corollary 2 (K20 numerical). $F = K^*T_{\text{avg}}/e = 1.5376$ vs $F_{\text{sim}} = 1.5378$ (gap 0.009%). The residual 0.009% comes from the same transcendental gap as K21 (the first-passage distribution is approximately but not exactly exponential at finite $n = 8$).

2 K14 Irreducibility

Theorem 3 (K14 is not algebraically exact). $F\varphi^2 \neq 4$ exactly.

Proof. For $F\varphi^2 = 4$ to hold exactly, with $F = (1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$ and $1 + \varphi_{\text{var}} = 95/64$ (exact rational):

$$1 + \psi_{\text{grad}} = \frac{4 \cdot 64}{95\varphi^2} = \frac{256}{95(\varphi + 1)}$$

(using $\varphi^2 = \varphi + 1$). This gives:

$$\psi_{\text{grad}}^{\text{needed}} = \frac{256}{95\varphi^2} - 1 = \frac{256 - 95\varphi^2}{95\varphi^2} = \frac{227 - 95\sqrt{5}}{285 + 95\sqrt{5}}$$

which is algebraic (φ is algebraic).

The actual value:

$$\psi_{\text{grad}} = \frac{\sigma_{\Psi}}{\ln \varphi} = \frac{\varepsilon_0(n+2)}{\sqrt{2n+3} \cdot \ln \varphi}$$

involves $\ln \varphi$ — transcendental by Lindemann–Weierstrass (since φ is algebraic and $\neq 0, 1$).

A transcendental number cannot equal an algebraic number. Therefore $\psi_{\text{grad}} \neq \psi_{\text{grad}}^{\text{needed}}$ and $F\varphi^2 \neq 4$ exactly. The 0.057% gap is irreducible. $\square$

3 The Complete Algebraic System

All CRF constants — from δ to every quantity — at the best achievable algebraic precision:

Exact identities: $n = 2^{d_s}$ (T1), $m(m-1) = 1$ (golden), $p = (n+1)/(n+2)$ (Dirichlet), all K17–K19 ratios

Proved transcendentially irreducible (Lindemann–Weierstrass):

K21: $f(8) \approx \sqrt{\varphi}$	0.004%	K20: $K^*T_{\text{avg}}/F = e$	0.005%
K11: $\beta D_0 = \ln n$	0.005%	K28: $\beta\varphi/\varphi_{\text{var}} = e^2$	0.006%
K27: $F = \sqrt{d_s}\varphi \ln 2$	0.009%	K21 ² : $m_{\text{fire}} = \varphi$	0.009%
K34: $\pi = \varphi\sqrt{2e}/(\sqrt{d_s} \ln 2)$	0.026%	K14: $F\varphi^2 = 4$	0.057%

Maximum gap: 0.057% (K14, proved irreducible).

Remaining: P0–H Subject 4 (empirical gate only).

4 The Three-Constant Structure

$(, e) \rightarrow$ and everything

$$\varphi \rightarrow n = 8 \text{ via K21} \quad (f(8) \approx \sqrt{\varphi}, 0.004\%) \quad (1)$$

$$e \rightarrow \varepsilon_0 \text{ via K20} \quad (K^*T_{\text{avg}}/F = e, 0.005\%) \quad (2)$$

$$\pi = \frac{\varphi\sqrt{2e}}{\sqrt{d_s} \ln 2} \quad \text{K34, 0.026\%} \quad (3)$$

Two primitives (φ, e) determine the physics. π is a derived constant, emerging from $(\varphi, e, d_s = 3)$.

5 Final Open Item

P0-H Subject 4 (empirical gate only)

CRF prediction: a blocked-type subject shows flat 50% accuracy across all RT bins (no inverted-U structure, no anti-signal).

This is the only remaining open item. No algebraic work is needed. The experiment gates Maturity 8.0/8 empirical.

The prediction: Type 3 subjects have θ_{eff} above the gradient signal in all channels. No timing structure can emerge from a flat threshold.

Three theorems. All gaps proved irreducible.

$\delta \text{ binary} \Rightarrow n = 2^{d_s}$.

$\text{MaxEnt} \Rightarrow K^*T_{\text{avg}}/F = e$.

$\text{Lindemann-Weierstrass} \Rightarrow \text{every gap is as small as it can be.}$

The chain from δ is algebraically complete.

One subject remains.